wet ion exchange resin. Following sorption (e.g. of M^+), the metal is eluted from the resin bed by passage of a solution (eluant) through the column. The eluant may contain a complexing anion or another metal ion (e.g. M^{3+}) which displaces M^+ through competition with it for positions on the resin. The metals are eluted by complexing agents in an order depending on their complex formation properties, as e.g. is illustrated for the lanthanides and actinides in Chapter 14 (Fig. 14.7).

In principle, *liquid partition chromatography* (LPC) is a liquid-liquid extraction where one of the liquid phases is stationary and attached to a supporting material, and the other liquid phase is mobile. It can be carried out with either the aqueous or the organic phase stationary; in the latter case the technique is referred to as reversed phase LPC. The aqueous phase can be made stationary by adsorption on silica gel, cellulose powder, etc. In order to make the organic phase stationary, beads (usually 50–200 μm) of PVC, teflon, Kel-F, etc., are being used.

Reversed phase LPC has been useful in radiochemistry for separating individual elements, e.g. lanthanides or actinides. It has also been used for separation of macro amounts of actinides. Instead of using columns in partition chromatography, a sheet of paper may be used to hold the stationary phase (*paper chromatography*) or an adsorbent coated on a glass plate (*thin-layer chromatography*). This technique has an advantage over column separations because the positions of the radioactive species are easily identified on the sheet, either simply by autoradiography (§18.4.4) or by scanning instruments. Paper chromatography is further described in an example in §18.4.1 and illustrated by Figures 18.7 and 18.8.

18.3. ANALYTICAL CHEMISTRY

18.3.1. Radiometric analysis

The term radiometric analysis is often used in a broad sense to include all methods of determination of concentrations using radioactive tracers. In a more restricted sense it refers to a specific analytical method which is based on a two-phase titration in the presence of a radioactive isotope. The endpoint of the titration is indicated by the disappearance of the radioisotope from one of the phases. Figure 18.4 illustrates two cases, (a) the determination of Ag^+ in a solution by titration with NaI solution containing $^{129}I^-$ ($\beta^-\gamma$ $t_{½}$ 1.57×10^7 y), and (b) the determination of Fe^{2+} in an aqueous solution, to which trace amounts of radioactive $^{55}Fe^{2+}$ (EC $t_{½}$ 2.73 y) has been added. In case (a) the AgI precipitate is radioactive but the solution has little radioactivity until all the Ag^+ has been precipitated. The activity of the solution is measured by a liquid flow GM-detector (Ch. 9). In the latter case (b) a two-phase liquid-liquid analytical technique is used (§18.2.6); the titrant contains a substance (oxine) which extracts Fe(II) from the aqueous to the chloroform phase. The radioactivity of the organic phase is followed by liquid scintillation (sampling) to determine the end point of the titration.

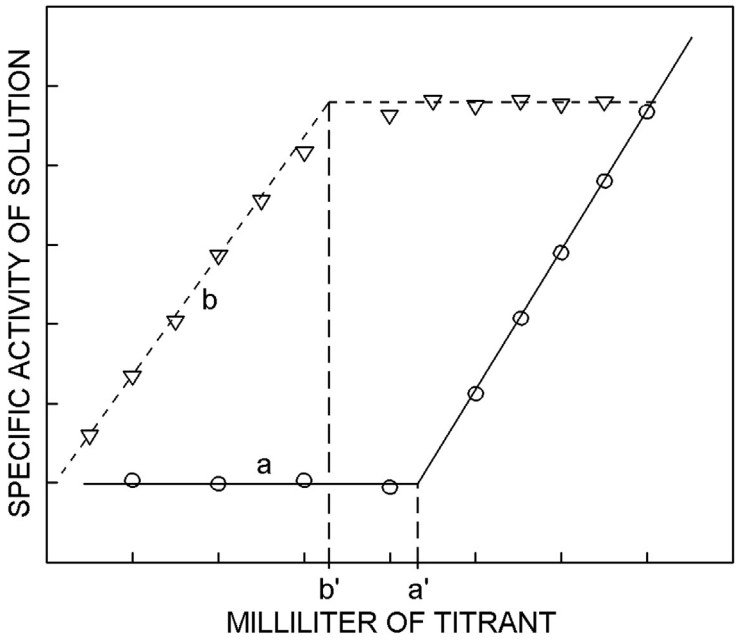


Figure 18.4 Radiometric titration curves.

Radiometric analysis is simple and rapid. Nevertheless, it is rarely used in analytical routine work, as a large number of multiple-element "instrumental" techniques are readily available (though the instruments usually are more expensive). Its most extensive use is for calibration of other techniques, and in analytical comparative techniques (e.g. environmental samples). In some cases results from different laboratories on common samples using various analytical techniques have differed by factors of 10^3 to 10^5! Because the radioactive tracer can be proven to be "atomically pure" (i.e. the radiation given off is unique for the radioisotope and element), it offers an absolute standard.

18.3.2. Isotope dilution analysis using radiotracers

In complex mixtures of compounds (for example, in organic synthesis or biochemical systems) it may be quite difficult to ascertain the exact amount of a specific component. A widely used technique of considerable value is *isotope dilution analysis*. This can be applied either with stable isotopes, in which case the detector is a mass spectrometer, or with radioactive isotopes, using measuring techniques presented in Chapter 9. The use of stable isotopes is usually limited to geologic samples, as described in §3.3, but may be applied to biological samples using highly enriched ^{13}C, ^{15}N or ^{18}O. However, the technique with radiotracers is more common due to its simplicity and lower cost.

A small radiochemically pure amount (w_0 g) of the selected compound ("reference") is added to the complex sample containing the unknown amount (w_u g) of the same compound. The reference may be either an element or a labelled compound, whose

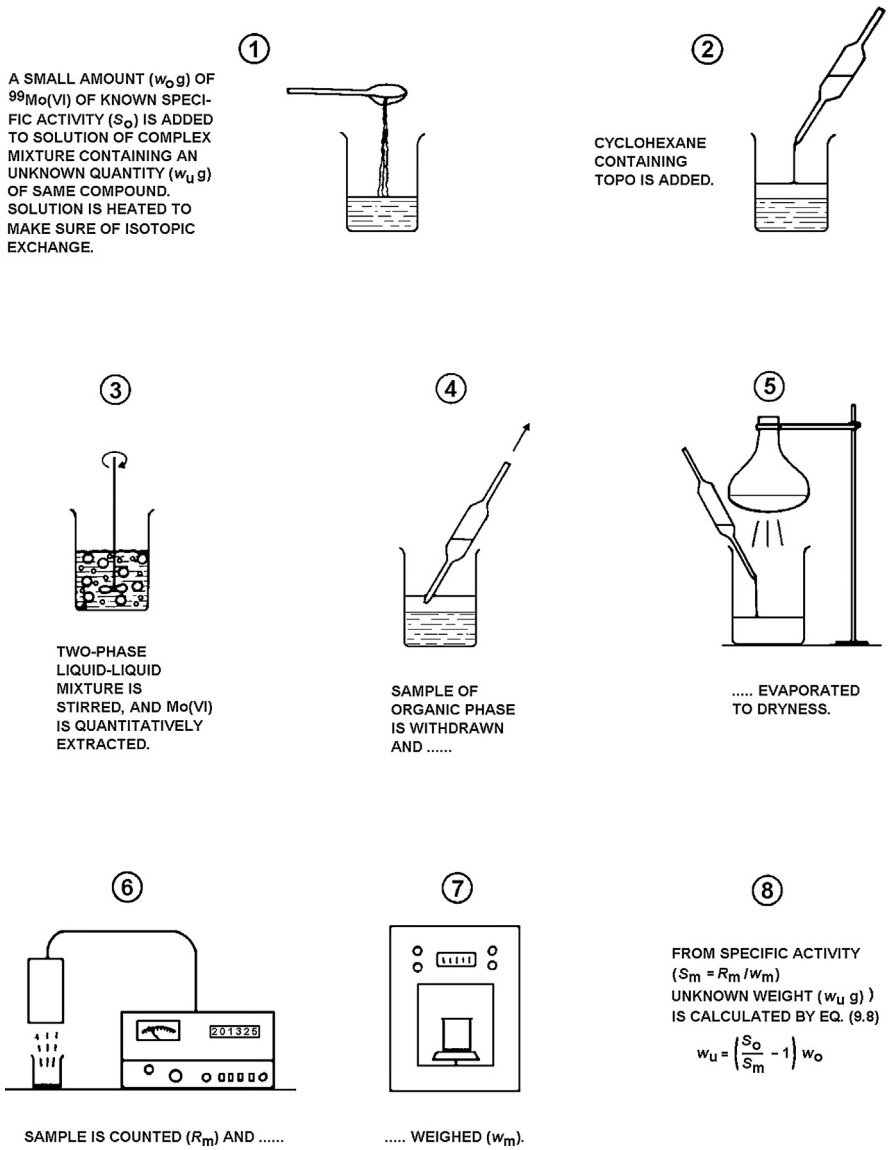


Figure 18.5 Sequence of steps in isotope dilution for determining the amount of a Mo(VI) complex in a composite mixture.

specific activity is known (S_0 Bq/g). After intimate mixing, the selected compound is isolated in high purity but not necessarily in high yield. The separated compound is weighed (w_m, g) and counted (R_m Bq) so that its specific activity (S_m, Bq/g) can be calculated. The method is pictured schematically in Figure 18.5. The weight, w_u, of the selected compound present in the original sample is calculated by

$$w_u = (S_0/S_m - 1)w_0 \tag{18.10}$$

The specific activity is defined by (5.49): $S = A/w\,\mathrm{Bq\,g^{-1}}$; since only the ratio S_0 / S_m is used in (18.10), the activity A can be replaced by the measured radioactivity R when the detection efficiency ψ is the same.

This technique is of particular advantage where quantitative separation of the desired compound is not feasible, as illustrated already by de Hevesy in 1932: In determination of micro amounts of lead by anodic precipitation, quite varying results were obtained. By addition of a known amount of "radiolead" and measuring the radioactivity of lead at the anode, the yield of the precipitation could be determined, and – although the electrolytic precipitation was inefficient – an exact analysis was obtained.

In some cases the measurement of the final sample utilizes a technique other than weighing, but the principle remains the same. Isotope dilution is used, for example, in the determination of naphthalene in tar, of fatty acids in mixtures of natural fat, of amino acids in biological material, etc.

18.3.3. Activation analysis

Activation analysis is a highly sensitive nondestructive technique for qualitative and quantitative determination of atomic composition of a sample. It has been particularly useful for determination of elements in complex samples (minerals, environmental samples, biological and archeological objects, etc.), because it provides a simple alternative to much more difficult, tedious and destructive techniques. Its main limitation is the demand for a strong irradiation source.

In activation analysis advantage is taken of the fact that the decay properties such as the half-life and the mode and energy of radioactive decay of a particular nuclide serve to identify uniquely that nuclide. The analysis is achieved by the formation of radioactivity through irradiation of the sample either by neutrons or charged particles. Neutron irradiation is by far the more common technique, and hence this method is often referred to as *neutron activation analysis*, NAA. A major advantage in activation analysis is that it can be used for the simultaneous determination of a number of elements and complex samples. If the counting analysis of the sample is conducted with a Ge-detector and a multichannel analyzer, as many as a dozen or more elements can be measured quantitatively and simultaneously (*instrumental NAA*, or *INAA*).

A sample is irradiated to form an amount R of radioactive nuclide according to the relationship (cf. §17.2):

$$R = \psi\,\varphi\,\sigma\,N\left(1 - \mathrm{e}^{-\lambda t_{\mathrm{irr}}}\right)\mathrm{e}^{-\lambda t_{\mathrm{cool}}} \tag{18.11}$$

We assume that irradiation is carried out by a homogeneous particle flux φ, in a neutron reactor. The minimum amount of an element which can be detected increases with the efficiency of the measuring apparatus ψ, the bombarding flux φ, the reaction cross-section σ, the irradiation time t_{irr} (up to saturation activity), the decay constant, λ, of the radioactive nuclide formed, and the time from end of bombardment to start of

Table 18.1 Limits of detection for 71 elements in a thermal neutron flux of 10^{17} n m^{-2} s^{-1} (1 h irradiation)

Limit of detection (μg)	Elements
$1-3 \times 10^{-6}$	Dy
$4-9 \times 10^{-6}$	Mn
$1-3 \times 10^{-5}$	Kr, Rh, In, Eu, Ho, Lu
$4-9 \times 10^{-5}$	V, Ag, Cs, Sm, Hf, Ir, Au
$1-3 \times 10^{-4}$	Sc, Br, Y, Ba, W, Re, Os, U
$4-9 \times 10^{-4}$	Na, Al, Cu, Ga, As, Sr, Pd, I, La, Er
$1-3 \times 10^{-3}$	Co, Ge, Nb, Ru, Cd, Sb, Te, Xe, Nd, Yb, Pt, Hg
$4-9 \times 10^{-3}$	Ar, Mo, Pr, Gd
$1-3 \times 10^{-2}$	Mg, Cl, Ti, Zn, Se, Sn, Ce, Tm, Ta, Th
$4-9 \times 10^{-2}$	K, Ni, Rb
$1-3 \times 10^{-1}$	F, Ne, Ca, Cr, Zr, Tb
10–30	Si, S, Fe

counting, t_{cool}. By proper selection of t_{irr} and t_{cool} the sensitivity for any element can be changed and interferences minimized. Table 18.1 shows the limits of detection in INAA.

Figure 18.6 shows a typical NAA spectrum obtained with a multichannel analyzer equipped with scintillation (upper curve) or semiconductor (lower curve) detectors. Each peak can be ascribed to a certain γ-energy, which in most cases identifies the nuclide. A number of nuclides can be identified simultaneously with semiconductor detectors, but with NaI(TI) scintillation detectors the poor resolution limits simultaneous multi-element analysis.

The area under the peak (shaded area in Fig. 9.5(b)) is proportional to the amount of the radioactive nuclide. If all other factors in (18.11) are known, the number of target nuclide atoms N can be calculated.

When complex mixtures are irradiated, such as geological or biological samples, there may be some difficulties in peak assignment. The energy spectrum is then scanned at repeated time intervals and from the decrease of the peak area with time the half-life of the peak may be established. This is a valuable additional aid in the assignment of the peak to a certain nuclide.

While in principle it is possible to calculate the amount of the desired element through the use of the proper values for the cross-section, flux, irradiation time, and half-life in (18.9), a simpler approach has been developed that avoids errors implicit in the uncertainties of each of these values. The unknown and a known standard of similar composition are irradiated and counted in an identical fashion. A direct comparison can be made according to the following relationship:

$$\frac{\text{Weight of element in unknown}}{\text{Weight of element in standard}} = \frac{\text{Activity of element in unknown}}{\text{Activity of element in standard}}$$

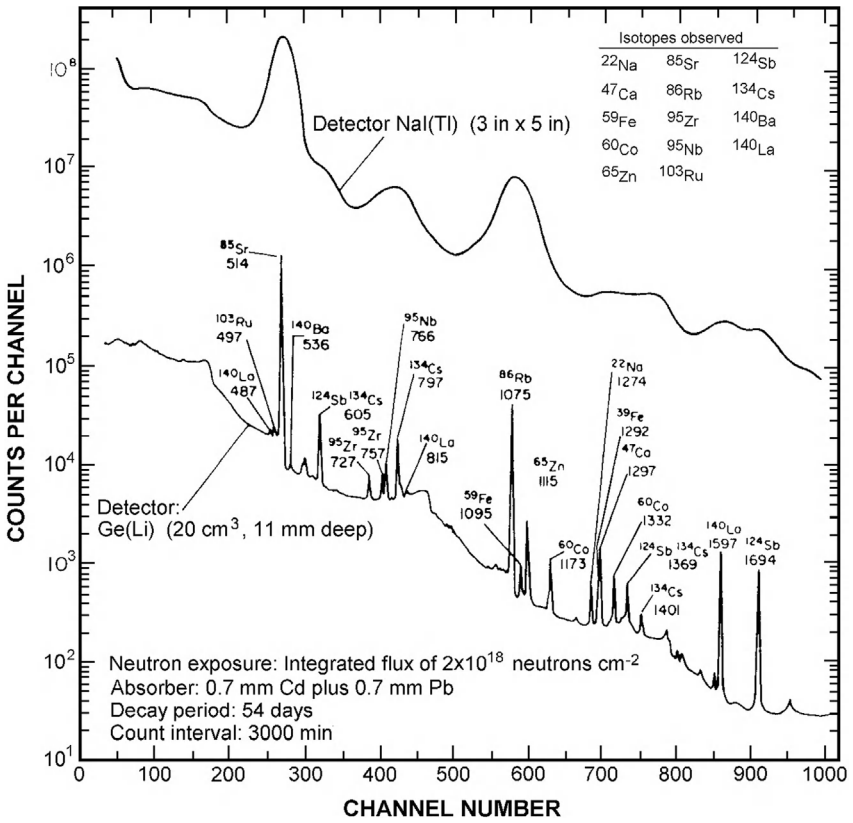


Figure 18.6 γ-spectrum of neutron-activated sea water. *(From Cooper, Wogman, Palmer and Perkins.)*

or

$$w_u/w_0 = R_u/R_0 \tag{18.12}$$

Sometimes such a large number of competing radioactivities are produced in the bombardment process that it may be necessary to conduct some chemical purification (RNAA, radiochemical NAA). This is particularly true if simple counting of β-activity or γ-ray spectrometry using NaI(TI) counting is used. However, with the development of semiconductor detectors and INAA the increased resolution in the spectrum allows simultaneous determination of as many as 15 or 20 competing radioactivities, usually without the necessity of chemical purification. Analysis of trace constituents in air and water, in soil and geological samples, in marine and in biological systems are some of the interesting applications of the NAA technique. Examples of on-line NAA include sorting ore minerals and oil well logging. In forensic science, by using NAA to measure the composition of the material adhering to a hand which has held a gun during firing, it

is possible to determine the type of ammunition and even the number of shots fired. Trace metal analysis of plants can be used to determine the location in which that plant has been grown (used, for example, for identification of marijuana growers). The trace constituents of archeological and art objects play an important role in ascertaining their authenticity and the identification of place of origin; the use of nondestructive NAA has been extremely valuable in this field. Activation analysis of the mineral content of pigments has enabled scientists to determine the authenticity of paintings attributed to certain artists since, in times past, each artist prepared his own paints by distinctive and individual formulae. A painting entitled "Christ and Magdalene" done in the Old Dutch style was proved to be a twentieth century forgery when NAA showed <7 ppm silver and <1.3 ppm antimony in the white lead paint. The sixteenth and seventeenth century Dutch paintings have white lead with about 10–1000 ppm silver and 50–230 ppm antimony.

It has been found that hair contains trace metals (e.g., Cu, Au, Ce, Na) in ratios which are typical for a particular individual, and activation analysis can be used to identify hair from a particular person. This application achieved public notice when it was found that hair from Napoleon had a relatively large amount of arsenic, indicating that some time prior to his death he had received large doses of arsenic. Through analysis of the hair of the Swedish king Erik XIV (who died suddenly in 1577 after a meal of pea-soup) it has been found that he must have received lethal amounts of arsenic as well as large amounts of mercury. The latter is assumed to have been taken into his body through the use of a mercury compound for treatment of an old wound.

The high sensitivity of activation analysis has made it very useful in environmental pollution studies. Table 18.2 lists the limits of detection for some elements in sea water under the conditions specified in the table.

Elements with very low sensitivity for thermal neutron bombardment (e.g. the lightest elements) can often be measured through irradiation with either fast neutrons (FNNA) or charged particles (CPAA); in the latter case eqn. (18.11) must be modified, see §17.2. Thus oxygen can be analyzed by bombardment with 14 MeV neutrons ($\sigma =$ 37 mb) yielding ^{16}N, which decays ($t_{½}$ 7.13 s) by emitting energetic β^- and γ (6–7 MeV). In FNAA and CPAA the flux may not be homogeneous, and (17.7)–(17.9) must be used. CPAA is usually employed for elements of atomic number less than 10 and is normally limited to surface analysis because of the short range of charged particles in solid materials. Surface concentrations of 0.01–0.02 μg/cm^2 can be detected. Irradiation by protons, deuterons, and ^{4}He have all been used. This technique should not be confused with PIXE described in §7.8.2. In PIXE high energy protons are used to induce X-ray fluorescence in the sample, and the analysis with the multichannel spectrometers is done simultaneously with the bombardment. In activation analysis the sample is counted after the end of the irradiation. Generally, charged particle irradiation techniques are more expensive than neutron activation analysis.

Table 18.2 Estimated minimum detectable concentrations of pollutant elements in sea water by INAA and by RNAA

Trace element	Typical reported concentrations in open ocean (μg l^{-1})	Minimum detectable concentrations (μg l^{-1})	
		INAA[(a)]	RNAA[(b)]
Hg	0.02—0.2	0.05	0.001
Cd	0.06—0.7	16000	0.001
Ag	0.002—0.05	1.0	0.003
As	2—3	Not possible	0.0001
Cu	0.5—2	Not possible	0.002
Cr	0.02—0.6	0.3	0.003
Zn	0.5—10	0.2	0.01
Sn	0.02	Not possible	9
Se	0.08	0.2	0.02
Sb	0.2	0.02	0. 00003

[(a)]25 ml sea water; 1 d irradiation at 10^{17} n m^{-2} s^{-1}; 40 d decay; 1000 min count on 20 cm^3 Ge(Li)-detector; based on 3 × above background-Compton contribution in peak areas.
[(b)]500 ml sea water; elements chemically separated; 1 d irradiation at 10^{17} n m^{-2} s^{-1}; 3 d decay; 500 min count on 20 cm^3 Ge(Li)-detector; based on 2 × above background-Compton contribution in peak areas.

18.3.4. Substoichiometric analysis

Isotope dilution analysis require the determination of either the chemical yield in the separation process or of the specific activity. This can be avoided by applying the *substoichiometric principle*, which may also increase the sensitivity of the analytical method.

The method requires three samples: (1) the test sample containing the unknown weight w_u of the element or compound X of interest; (2) a standard of the same element (preferably in a similar matrix, subscript s); and (3) a nonactive carrier (usually a solution, subscript c), of the same element or compound X.

If a test sample containing the unknown weight w_u of the element or compound X and a standard containing a known weight w_s^0 of the same are irradiated under identical conditions,

$$w_u = w_s^0 A_u / A_s \tag{18.13}$$

where A_u and A_s (or R_u and R_s, if measured under identical conditions) are the total radioactivities contained in sample and standard. If the test sample contains other radioactivities in addition to that in the standard, the preceding relation can be applied only when the measurement technique is isotope specific, for example, when high-resolution gamma spectroscopy is used, in which case interference from undesirable radioactivity is eliminated.

When simple counting equipment must be used or highly complicated test samples are involved, it may be desirable to isolate chemically the element or compound of

interest. This can be done through selective chemical procedures using a *non-radioactive carrier* of element X. The carrier is added to the irradiated test sample (containing w_u), which is processed through the different purification steps. Finally a sample of almost (or sufficiently) pure X is isolated, and its activity R_{u+c} and weight w_{u+c} measured. It is also necessary to run the irradiated standard with added carrier through the same chemical separation procedures. With the exception of w_u, the final equation (18.14) has terms that are known or measured:

$$w_u = w_s^0 \left\{ R_{u+c} \left(w_c^0 + w_u \right) / w_{u+c} \right\} / \left\{ R_{s+c} \left(w_{s+c}^0 + w_s^0 \right) / w_{s+c} \right\} \tag{18.14}$$

Usually, the amount of carrier is much greater than the unknown and the standard, that is, $w_c^0 \gg w_s^0$, and $\gg w_u^0$. Then if equal amounts (by weight) of the nonactive isotopic carrier are added to the unknown sample and the standard, the equation reduces to

$$w_u = w_s^0 \, (R_{u+c} \, w_{s+c}) / (R_{s+c} \, w_{u+c}) \tag{18.15}$$

This equation can be simplified even further. In the chemical separation procedure a *substoichiometric amount* of reagent is added, provided the conditions are such that this reagent quantitatively reacts with X. For example, zinc is extracted quantitatively from aqueous solutions buffered at pH 8 by dithizone in chloroform. If the amount of dithizone is less than that of zinc after adding a carrier of nonradioactive $ZnSO_4$, only part of the zinc (e.g., 25%) is extracted, but the dithizone is quantitatively bound to zinc in the organic phase. Thus varying amounts of zinc in the aqueous phase do not change the amount in the organic phase, which is constant, though the specific activity varies. The two chloroform solutions, from test plus carrier, and from standard plus carrier, contain equal amounts of Zn. Liquid-liquid extraction is commonly used for this technique.

The foregoing is an example of the basic principle of substoichiometric analysis. Under these conditions $w_{s+c} = w_{u+c}$ and

$$w_u = w_s^0 R_{u+c} / R_{s+c} \tag{18.16}$$

Thus by carefully choosing proper experimental conditions, the analytical procedure is reduced to two radioactivity measurements. For precise results the value of w_u must be comparable to w_s^0, which can be ascertained by a few initial survey experiments.

This substoichiometric analysis technique can be applied to most metals with a high degree of accuracy and a sensitivity of 10^{-6} to 10^{-10} g of X.

When the substoichiometric principle is applied to isotope dilution analysis, the relationship becomes

$$w_u = w_s^0 \, (R_s / R_{u+s} - 1) \tag{18.17}$$

where w_s^0 is the weight of the standard added, and R_s and R_{u+s} the radioactivities measured from the substoichiometrically separated samples of the standard and of the mixture of standard and unknown. The specific activities need not be determined as in (18.8) because equal weights of standard (w_s) and sample plus standard (w_{u+s}) are isolated by using the substoichiometric principle.

The usefulness of this technique for routine determination of numerous chemical elements in various matrixes has been demonstrated by Ruzicka, Stary, and others. It is also applicable to organic compounds and known in medicine as radioimmunoassay.

18.4. APPLICATIONS TO GENERAL CHEMISTRY

In no other area have radioactive tracers played such an important role as in the studies of chemical and biological reaction paths. This is, of course, due to that, in principle, in each radioactive decay the atom announces its position. Thus the detection sensitivity can approach the ultimate limit. A radioactive nuclide, as for example ^{14}C (β^- $t_{½}$ 5730 y), ^{32}P (β^- $t_{½}$ 14.282 d), or ^{198}Au ($\beta^-\gamma$, $t_{½}$ 2.6935 d), can be followed through a number of different chemical reaction steps, revealing details of metabolic or process reactions impossible to discover by other techniques.

18.4.1. Determination of chemical reaction paths

The use of radioisotopes in the study of the steps in a chemical reaction system is well established. Let us consider a few examples to illustrate this technique.

If phenol is labelled with deuterium or tritium in the hydroxyl group and warmed to a temperature slightly below decomposition, the labelled hydrogen migrates to other hydrogen positions of the benzene ring either by intra-molecular rearrangement or by inter-molecular reactions. However, using C_6H_5OT and C_6H_4TOH yields C_6H_4TOT, which can be formed only through inter-molecular collisions, thus eliminating intra-molecular rearrangement as the reaction mechanism.

The study of the reaction steps in the photosynthesis of carbohydrates from atmospheric CO_2 in the presence of light and chlorophyll is an outstanding example of the value of the tracer technique. The overall process (which involves many steps) can be written as

$$6\ CO_2 + 12\ H_2O \xrightarrow[\text{chlorophyll}]{\text{light}} C_6H_{12}O_6 + 6\ O_2 + 6\ H_2O \qquad (18.18)$$

Using partition chromatographic technique and tracers of ^{14}C, ^{32}P, and T, Calvin and co-workers were able to identify the intermediate steps involved. The experimental procedure is usually as follows (Fig. 18.7). Plants are placed in atmospheres containing ^{14}C-labelled CO_2 and irradiated with light. After different irradiation times the plants are removed and sections are digested to dissolve the material. A few drops of the solution